

Low Quiescent, High Efficiency 1.5 A Buck-Boost Converter with I²C Interface

Features

- Buck, Buck-Boost and Boost Operation with Seamless Mode Transition
- 2.2V to 5.5V Input Supply Voltage Range
- 2.025V to 5.2V Output Voltage Range with Digitally Programmable (25mV/steps)
- 2.8uA Low Quiescent Current
- 0.3uA Shutdown Current
- Excellent Load Transient Response
- Allows Dynamically-Voltage-Scaling Control
- Automatic PFM Mode and Forced PWM Mode Selection
- Up to 1MHz I²C Interface(I²C supports 1.2V I/O)
- Output Voltage Selection
(VSEL=H, VOUT=3.5V Default; VSEL=L, VOUT=3.3V Default)
- Maximum Continuous Output Current:
Up to 0.8A for VIN ≥ 2.2V, VOUT = 3.5V
Up to 1A for VIN ≥ 2.5V, VOUT = 3.5V
Up to 1.5A for VIN ≥ 3.3V, VOUT = 3.5V
- WLCSP 1.3X1.3-9B Package

Applications

- Wearable Devices
- Portable Devices
- TWS Earbud Chargers
- Optical Heart Rate Monitor LED Bias
- Battery Powered Systems
- Smartphones

General Description

The AWP37701Z is a high-efficiency, single inductor, advanced COT synchronous Buck-Boost converter with 2.2V to 5.5V wide input voltage range and well regulate to the digitally programmable output voltage from 2.025V to 5.2V. Which is suitable for wide input supply range applications, regardless of input voltage is lower, higher than or even equal to the output voltage. The COT control architecture features outstanding line/load transient response, seamless transition between buck and boost modes, provides stable operation with small ceramic output capacitors and without complicated external compensation.

The AWP37701Z features I²C interface, which allows programmable output voltage, soft-start slew-rate adjusted and device status monitoring. The target output voltage can also be switched through external VSEL pin to perform dynamically-voltage-scaling (DVS), and the ramp-up slew-rate and ramp mode of DVS can also be set by setting the related registers.

The AWP37701Z has internal soft start module to limits the inrush current. Full protection features include over current protection(OCP), short circuit protection(SCP), over voltage protection(OVP), under voltage protection(UVP) and over temperature protection(OTP). The AWP37701Z is available in WLCSP 1.3X1.3-9B package.

Typical Application Circuit

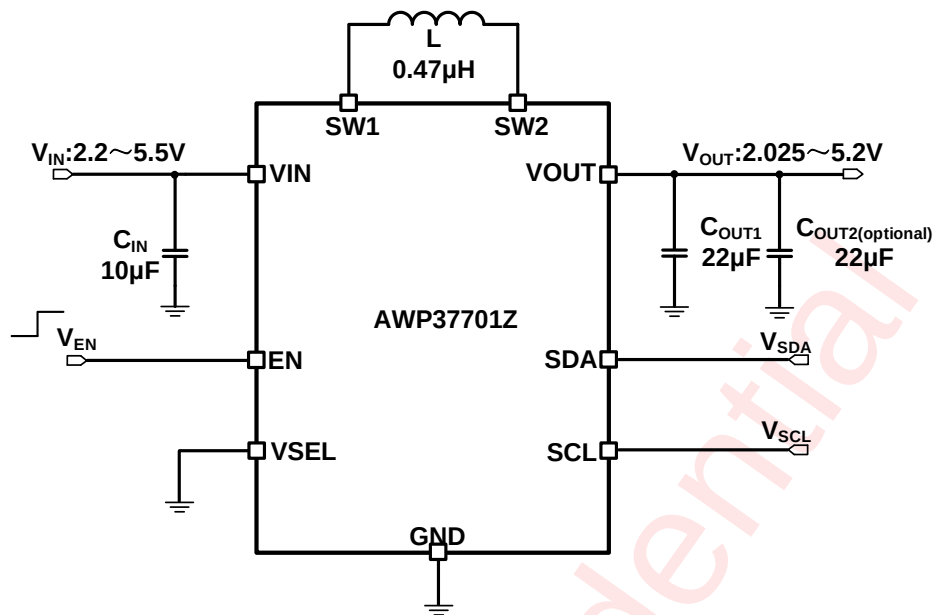


Figure 1 AWP37701Z Typical Application Circuit

NOTE: When the output current is $\leq 1A$, a 22µF C_{OUT} is recommended. When the output current is $>1A$, $2 \times 22\mu F$ C_{OUT} capacitors are recommended. The recommended maximum output current should not exceed 1.5A.

Pin Configuration And Top Mark

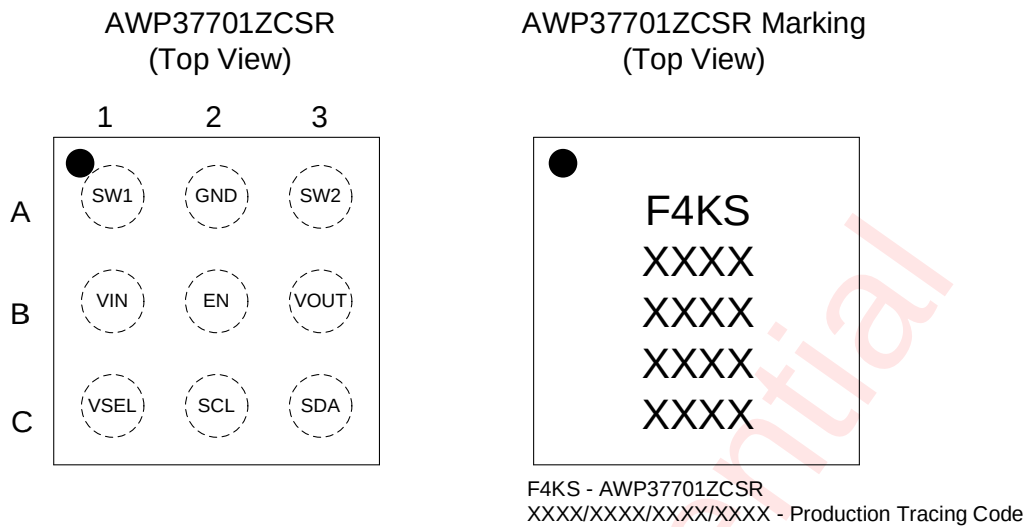


Figure 2 Pin Configuration and Top Mark

Pin Definition

No.	NAME	DESCRIPTION
A1	SW1	Switching node 1. Connect to the inductor.
A2	GND	Ground. All signals are referenced to this pin.
A3	SW2	Switching node 2. Connect to the inductor.
B1	VIN	Power input.
B2	EN	Enable control input. A logic-high enables the converter; a logic-low forces the device into shutdown mode. Floating is prohibited.
B3	VOUT	Power output.
C1	VSEL	Voltage select pin. When this pin is logic low, VOUT is set by the VOUT1 register; when this pin is logic high, VOUT is set by the VOUT2 register. The VSEL pin shall be tied high or pulled to GND. Floating is prohibited.
C2	SCL	I ² C serial interface clock.
C3	SDA	I ² C serial interface data.

Functional Block Diagram

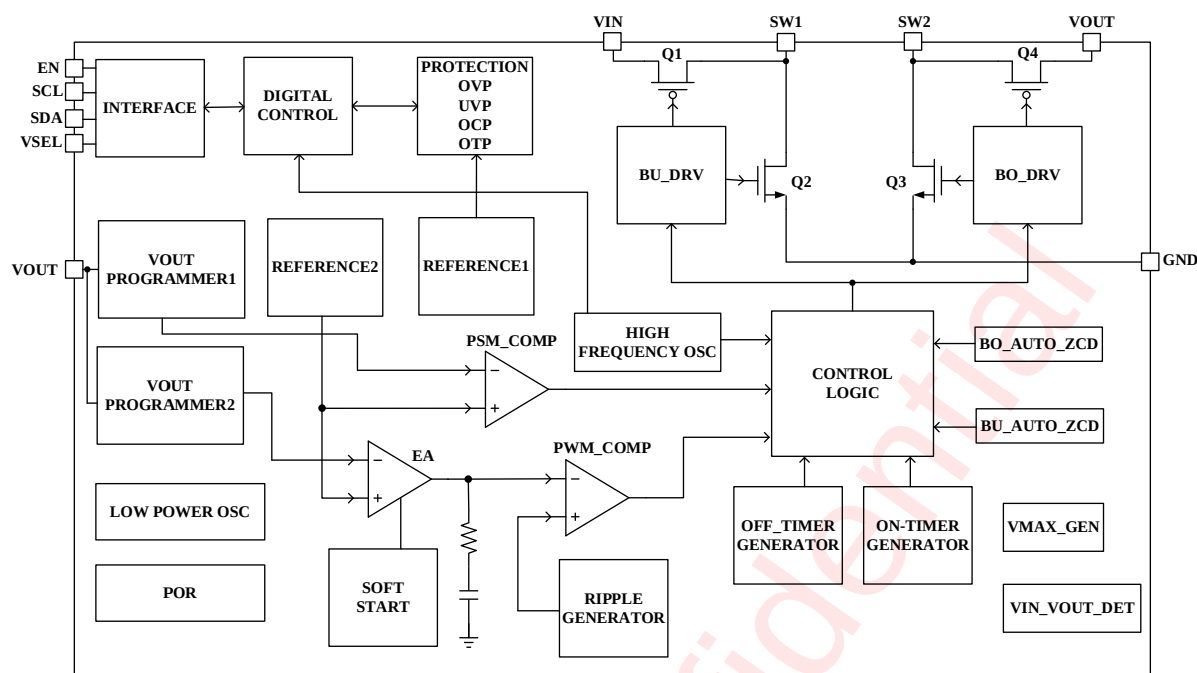


Figure 3 Functional Block Diagram

Ordering Information

Part Number	Temperature	Package	Marking	Moisture Sensitivity Level	Environmental Information	Delivery Form
AWP37701ZCSR	-40℃~85℃	WLCSP 1.3X1.3-9B	F4KS	MSL1	ROHS+HF	4500 units/ Tape and Reel

Absolute Maximum Ratings^(NOTE1)

PARAMETERS	RANGE
Supply voltage range VIN	-0.3V to 6V
Power output voltage range VOUT	-0.3V to 6V
SW1、SW2	-0.3V to 6.5V
IO voltage range EN、VSEL、SDA、SCL	-0.3V to 6V
Junction-to-ambient thermal resistance θ_{JA} ^(NOTE2)	100.3°C /W
Maximum operating junction temperature T _{JMAX}	150°C
Storage temperature T _{STG}	-65°C to 150°C
Lead temperature (soldering 10 seconds)	260°C
ESD(Including CDM ^(NOTE 3) HBM ^(NOTE 4))	
CDM	±1.5kV
HBM	±2kV
Latch-Up	
Test condition: JESD78F	+IT: 200mA -IT: -200mA

NOTE1: Conditions out of those ranges listed in "absolute maximum ratings" may cause permanent damages to the device. In spite of the limits above, functional operation conditions of the device should within the ranges listed in "recommended operating conditions". Exposure to absolute-maximum-rated conditions for prolonged periods may affect device reliability.

NOTE2: Based on simulations conducted with the device mounted on a JEDEC 4-layer PCB.

NOTE3: Test method: ESDA/JEDEC JS-002-2022;

NOTE4: The human body model is a 100pF capacitor discharged through a 1.5kΩ resistor into each pin. Test method: ESDA/JEDEC JS-001-2023;

Recommended Operating Conditions

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
VIN	Input voltage		2.2		5.5	V
VOUT	Output voltage		2.025		5.2	V
IOUT	Output current	Vout=3.5V, VIN ≥ 2.2V	0.8			A
		Vout=3.5V, VIN ≥ 2.5V	1			A
		Vout=3.5V, VIN ≥ 3.3V	1.5			A
L	Inductor		0.22	0.47	1	μH
CIN	Input capacitor			10		μF
COUT	Output capacitor			22		μF
TA	Operating free-air temperature range		-40		85	°C

Note: All the input and output capacitors are the suggested values, referring to the effective capacitances, subject to any de-rating effect, like a DC bias.

Electrical Characteristics

$V_{IN}=3.8V$, $V_{OUT}=3.5V$, $T_A=25^{\circ}C$, $C_{OUT} = 22\mu F$, $L = 0.47\mu H$, for typical values (unless otherwise noted)

PARAMETER		TEST CONDITION	MIN	TYP	MAX	UNIT
V_{IN}	Input Operating Voltage		2.2		5.5	V
V_{UVLO}	UVLO Threshold Voltage	V_{IN} rising		1.92	2.05	V
	Hysteresis For UVLO			80	200	mV
I_{Q_SW}	Quiescent Current	$V_{EN}=3.8V, I_{OUT}=0A$		2.8	5	μA
$I_{Q_NON\ SW}$	Quiescent Current	$V_{EN}=3.8V, V_{OUT}=3.8V$, no switching		2.3	5	μA
I_{SD}	Shutdown Current	$V_{EN}=0V$		0.3	1	μA
V_{OUT}	Output Voltage Range		2.025		5.2	V
I_{OUT_MAX}	Maximum Continuous Output Current	$V_{IN}=2.2 - 2.5V, V_{OUT}=3.5V$	0.8			A
		$V_{IN}\geq 2.5V, V_{OUT}=3.5V$	1			A
		$V_{IN}\geq 3.3V, V_{OUT}=3.5V$	1.5			A
$V_{OUTVSEL_L}$	Default Output Voltage	$V_{SEL} = low$		3.3		V
$V_{OUTVSEL_H}$	Default Output Voltage	$V_{SEL} = high$		3.5		V
R_{DISCHG}	Output Discharge Resistor	$V_{EN} = 0V$		200		Ω
$A_{CC_Forced}^{NO\ TE1}$	Output Voltage Accuracy	Forced PWM operation	-1		1	%
$A_{CC_AUTO}^{NOT\ E1}$	Output Voltage Accuracy	Auto PFM operation	-1		3	%
I_{LIM_BUCK}	MOSFET Peak Current Limit	Buck operation		3.7		A
I_{LIM_BOOST}	MOSFET Peak Current Limit	Boost operation		3		A
F_{SW}	Switching Frequency	Boost or Buck operation	1	1.6		MHz
V_{UVP}	Positive-Going Under-Voltage Threshold Voltage		92	95	98.5	%
	Negative-Going Under-Voltage Threshold Voltage		82	85	88	%
t_{SS}	Soft Start Time				1	ms
t_{EN}	Enable Delay Time			130		μs
D_{VS_SR}	Output Voltage Dynamic Voltage Scaling Slew Rate	0x01, bit[1:0] = 00b		1		V/ms
		0x01, bit[1:0] = 01b		2.5		V/ms
		0x01, bit[1:0] = 10b		5		V/ms

		0x01, bit[1:0] = 11b		10		V/ms
$R_{DS(on)}_{Q1}$	Buck High-Side Switch On Resistance(Q1)			50		mΩ
$R_{DS(on)}_{Q2}$	Buck Low-Side Switch On Resistance(Q2)			50		mΩ
$R_{DS(on)}_{Q3}$	Boost Low-Side Switch On Resistance(Q3)			40		mΩ
$R_{DS(on)}_{Q4}$	Boost High-Side Switch On Resistance(Q4)			60		mΩ
I_{IH}	High-Level Input Current	$V_{SCL} = V_{SDA} = V_{SEL} = 3.3V$, no pull-up resistor			1	μA
I_B	Input Bias Current	$V_{EN} = 3.8V$			1	μA
I_{HS_LK}	High-Side Switch Leakage Current	$V_{EN} = 0V$, $V_{SW} = 0V$			1	μA
V_{EN_H}	Enable Input High Threshold	$V_{IN} = 2.2V$ to $5.5V$	1.2			V
V_{EN_L}	EN Falling Threshold	$V_{IN} = 2.2V$ to $5.5V$			0.4	V
V_{IH}	(SCL, SDA, VSEL) Input High Threshold		0.84			V
V_{IL}	(SCL, SDA, VSEL) Input Low Threshold				0.36	V
T_{OTP}	Over Temperature Protection			145		°C
	Over Temperature Hysteresis			20		°C

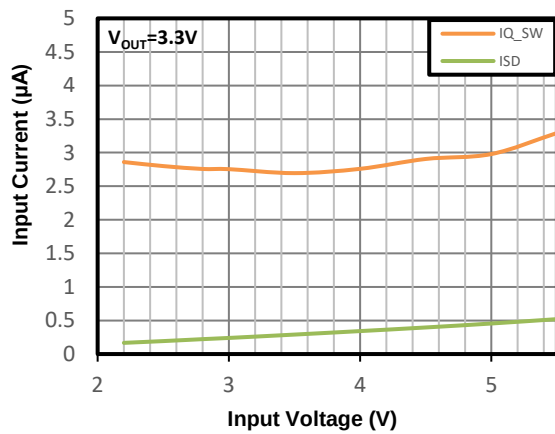
Note1: The minimum and/or maximum limits are guaranteed via production testing (under a subset of conditions) and/or characterization. Full coverage across all specified conditions is not achieved through production testing alone.

I²C Characteristics

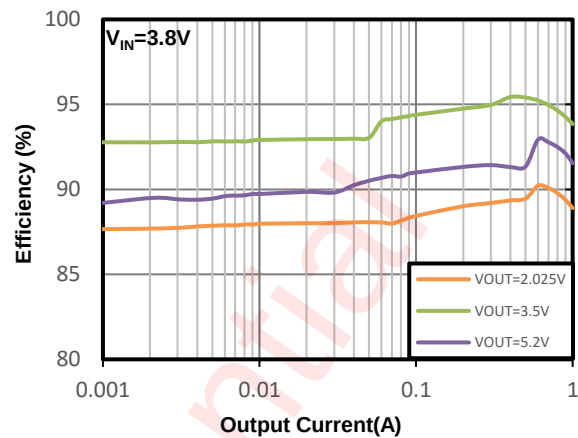
PARAMETER		TEST CONDITION	MIN	TYP	MAX	UNIT
V ^{I²C} OL	Logic Output Threshold Voltage (SCL, SDA, VSEL)				0.4	V
V ^{I²C} int	I ² C Work Voltage			1.8		μA
I _{IN,I²C}	Input Current Each IO Pin		-10		10	μA
t _{DS,I²C}	Data Set-Up Time		70			ns
f _{CLK}	SCL Clock Frequency		100	400	1000	kHz
t _{BUF}	Bus Free Time between Stop and Start Condition		0.5			μs
t _{HD,STA}	Hold Time (Repeated) START Condition		0.26			μs
t _{SU,STA}	Set-Up Time for a Repeated START Condition		0.26			μs
t _{HD,DAT}	Data Hold Time		0.1			ns
t _{SU,STO}	Set-Up Time for STOP Condition		0.26			μs
t _{VD,ACK}	Data Valid Acknowledge Time				0.45	μs
t _{SU,DAT}	SDA Set-Up Time		50			ns
t _{LOW}	Low Period of the SCL Clock		0.5			μs
t _{HIGH}	High Period of the SCL Clock		0.26			μs

Typical Characteristics

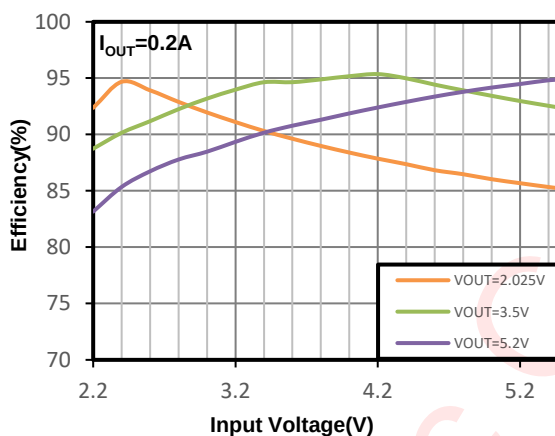
$T_A=25^{\circ}\text{C}$, $V_{IN}=3.8\text{V}$, $C_{OUT}=22\mu\text{F}$, $L=0.47\mu\text{H}$, Auto PFM mode, unless otherwise noted .



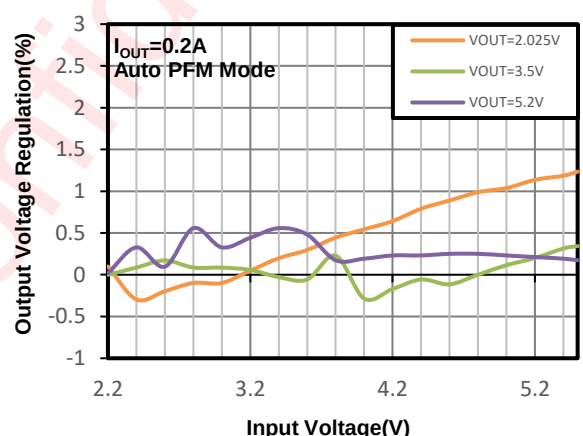
Input Current vs Input Voltage



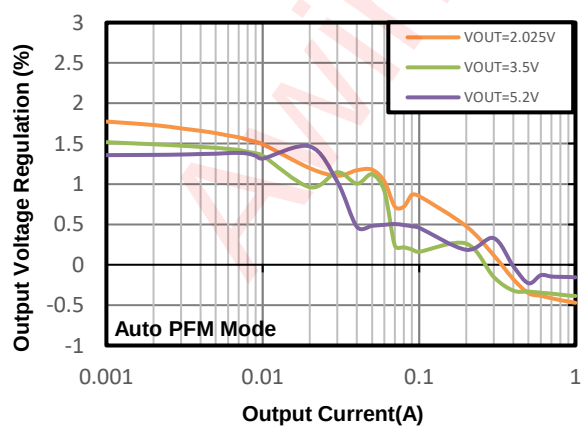
Efficiency vs Output Current



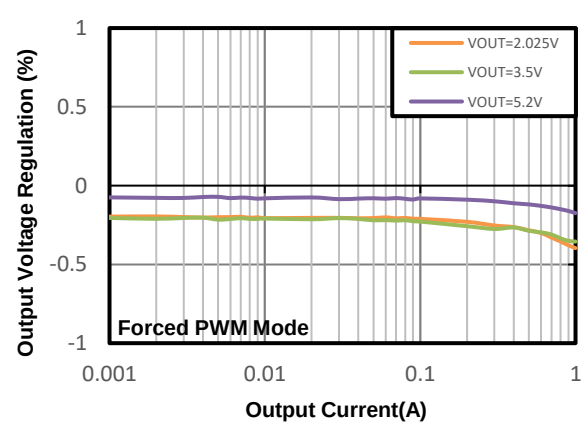
Efficiency vs Input Voltage



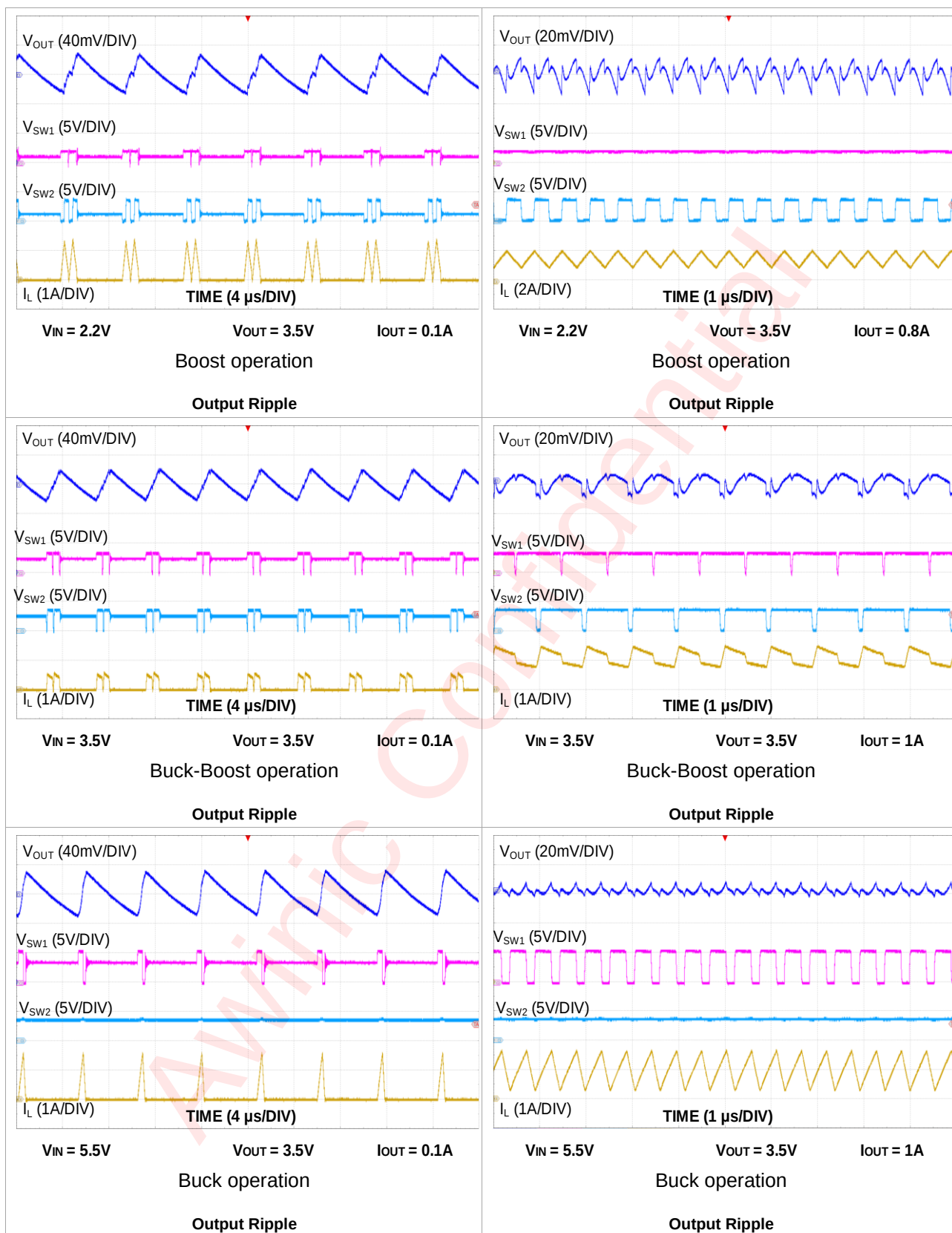
Output Voltage Regulation vs Input Voltage

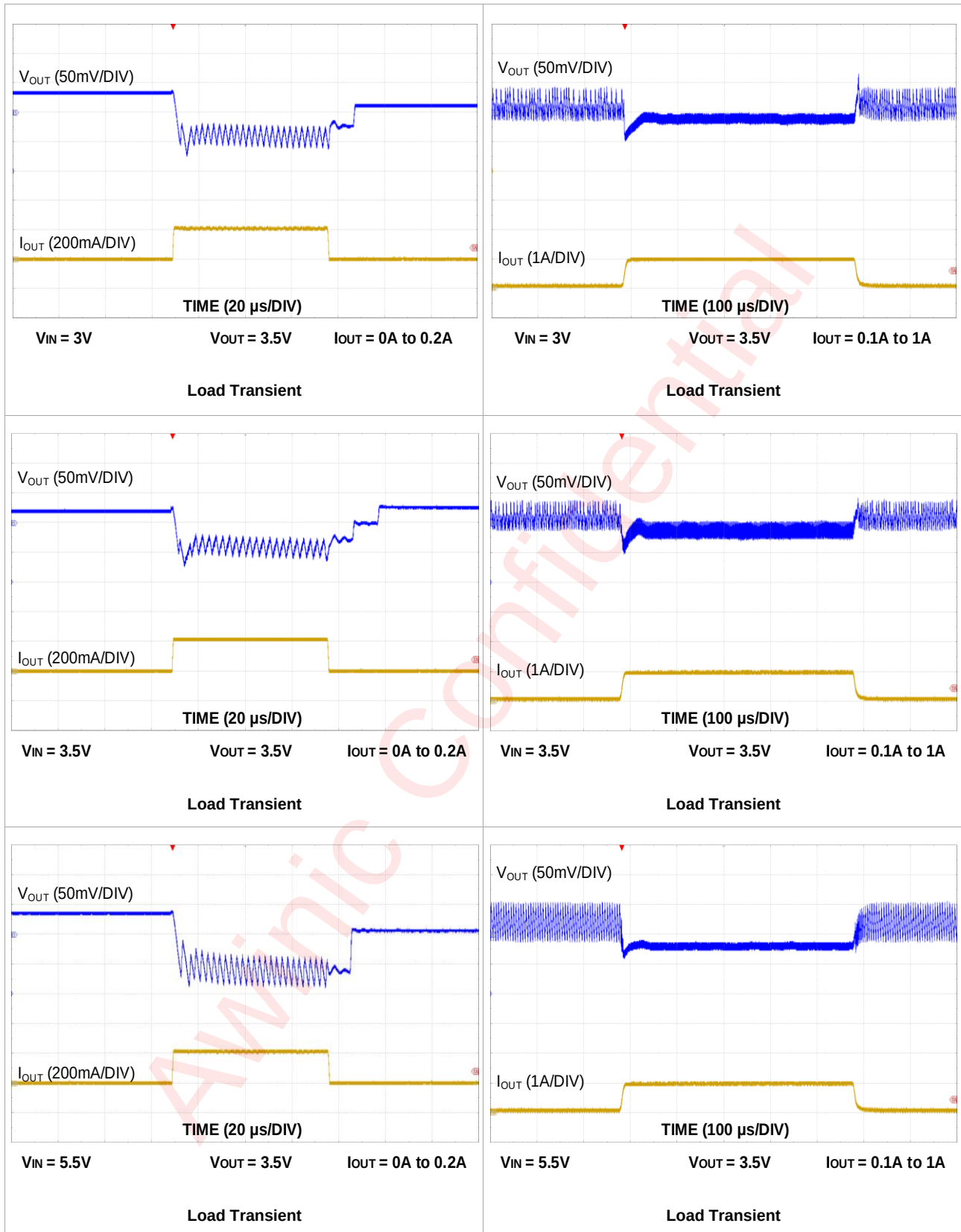


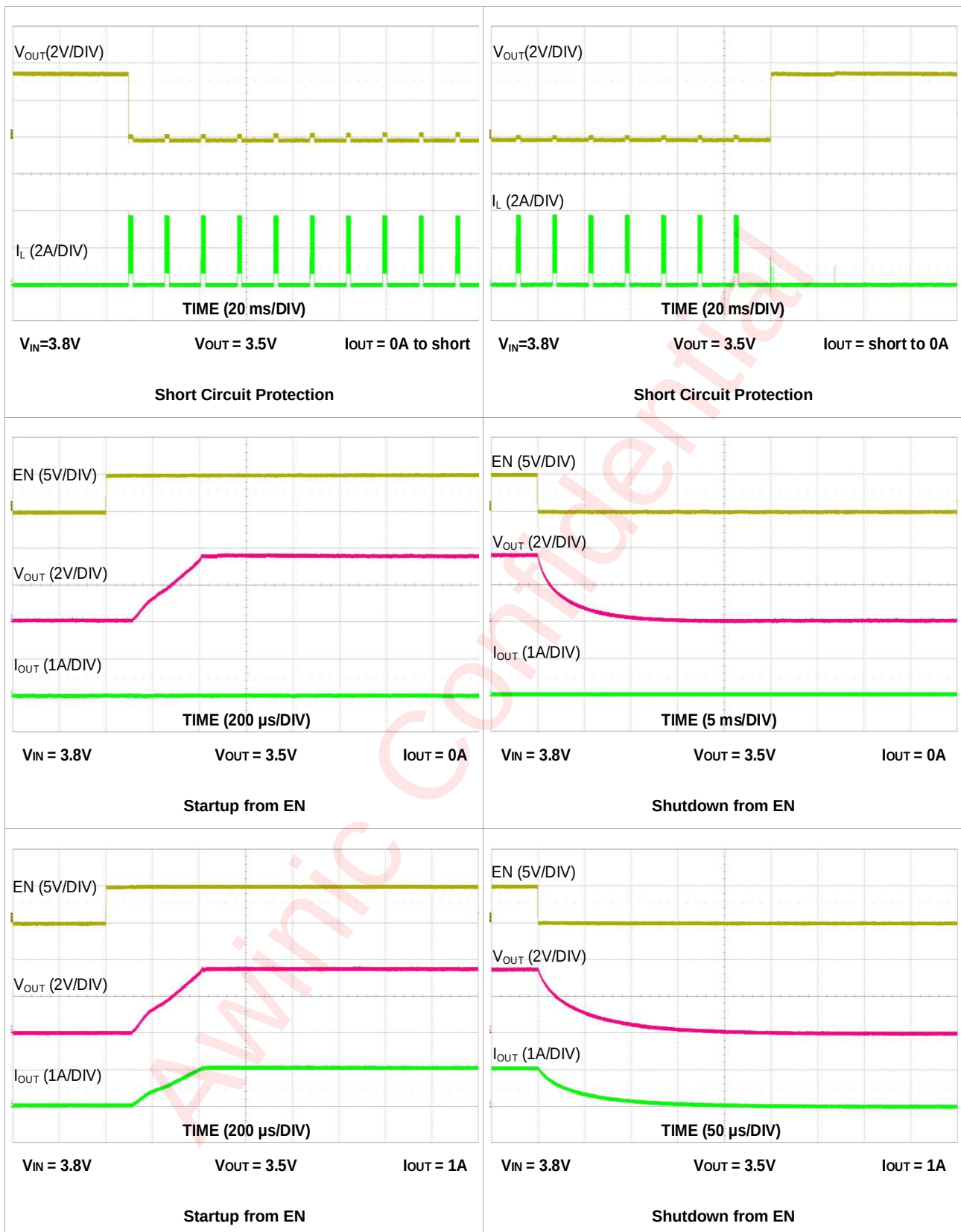
Output Voltage Regulation vs Output Current

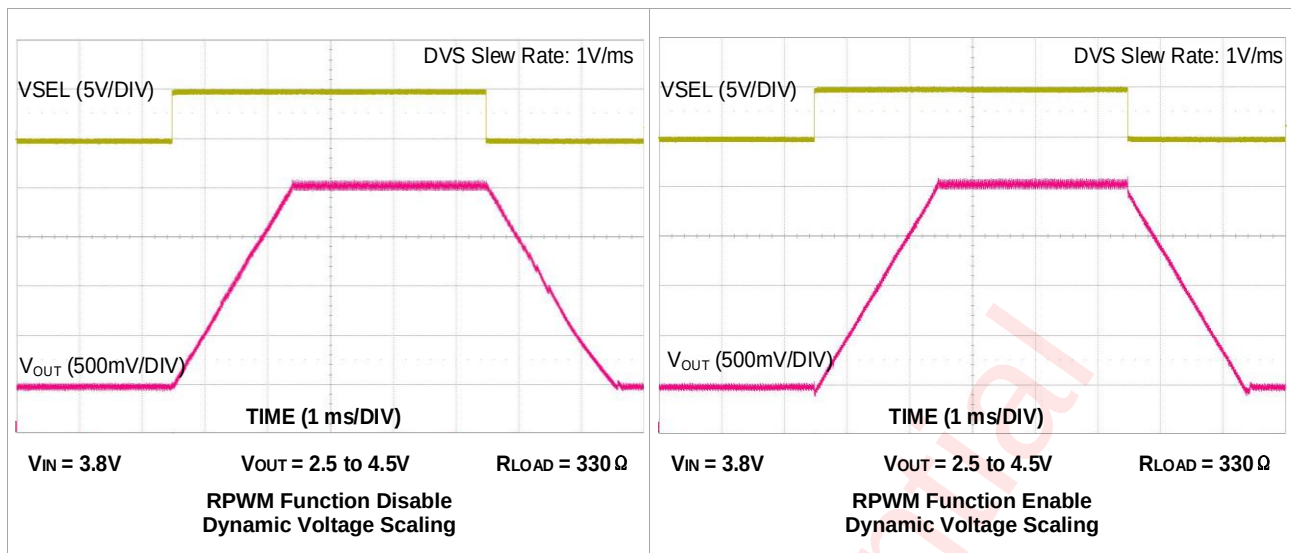


Output Voltage Regulation vs Output Current









Detailed Functional Description

Control Scheme

AWP37701Z uses emulated current ripple constant on-time control scheme where an outer voltage control loop generates the demand signal for an inner emulated current control loop. During the off-time, the control loop monitors the emulated current ripple signal with demand signal, and when the emulated current signal equals the demand signal from the error amplifier, the off-time stops and the next on-time of the switching cycle starts. The on-time is a function of VIN and VOUT. In this way, a pseudo-fixed switching frequency can be achieved for different input and output voltages.

This control method results in excellent dynamic response for the chip. For instance, when VIN=3.8V, VOUT=3.5V, and IOUT ranges from 0 to 0.3A, the chip maintains stable performance under repeated load draws. Specifically, after applying the load for 10 μ s, The difference between the output voltage's persistence and ripple remains within 10 mV. This indicates the chip's ability to efficiently handle rapid changes in load current and maintain voltage stability, which is crucial for many electronic applications requiring consistent power supply performance.

Operation Mode

AWP37701Z automatically switches between buck, boost, and buck-boost operation as required by the VIN and VOUT operating conditions. The transition between different operating mode (buck, boost, or buck-boost) of the converter is automatic and seamless. AWP37701Z operates as a true buck converter when VIN > VOUT. AWP37701Z operates as a true boost converter when VIN < VOUT. When VIN and VOUT gets closer the device operates in buck-boost mode.

Soft-Start

An internal current source charges an internal capacitor to build the soft-start ramp voltage. During the soft-start period, PG is initially set to '0'. and when VOUT reaches 95% of its set voltage, the device sets PG to '1'. The rise time of the output voltage changes with the application circuit and the operating conditions. The Enable Delay Time is set to 0.13mS, the soft-start time is set to 0.25mS. The output voltage rise time increases if the load current is large or the output capacitance is large.

VSEL

VSEL = L output voltage default setting is 3.3V that can be programmed by Address 0x04[6:0] VOUT1.

VSEL = H output voltage default setting is 3.5V that can be programmed by Address 0x05[6:0] VOUT2.

VSEL = L the device address should be configured to 76h (1110110b).

VSEL = H the device address should be configured to 75h (1110101b).

The VSEL pin shall be tied high or pulled to GND. Floating is prohibited.

Auto PFM (Pulse Frequency Modulation) Mode

In order to save power and improve efficiency at low loads, the Buck/Boost operate in PFM (Pulse Frequency Modulation) mode to save power. During PFM mode, the switching frequency is proportional to loading to reach output voltage regulation. When load increases and inductor current becomes continuous again, the Buck/Boost automatically goes back to PWM fixed frequency mode.

Additionally, AWP37701Z will enter Deep Sleep Mode to achieve extremely low quiescent current when there's no load. (see Deep Sleep Mode for more information).

Deep Sleep Mode

AWP37701Z is suitable for low quiescent current applications. When there's no load, AWP37701Z will automatic enter Deep Sleep Mode to achieve extremely low quiescent current. Once the load increases, it will go back to normal operation at once.

FPWM (Forced Pulse Width Modulation) Mode

The switching frequency is forced into PWM mode operation. In this mode, the inductor current is in CCM (Continuous Current Mode) and the voltage is regulated by PWM control.

To enable Forced-PWM operation, set the FPWM bit (Address 0X01[3]) to '1'.

VOUT Program

The AWP37701Z supports programmable VOUT from 2.025V to 5.2V with 25mV resolution. The output voltage can be set by register bit and the output voltage is given by the following equation:

$$VOUT = 2.025V + VOUTreg \times 25mV$$

For example:

If VOUTreg = 110011 binary (51 decimal), then $VOUT = 2.025V + 51 \times 25mV = 2.025V + 1.275V = 3.3V$.

AWP37701Z also has external VSEL pin to select VOUT1 (Address 0X04[6:0]) or VOUT2 (Address 0X05[6:0]). Pulling VSEL to high is for VOUT2 and pulling VSEL to low is for VOUT1. Upon POR, VOUT1 and VOUT2 are reset to their default voltages.

Dynamically Voltage Scaling Control

The AWP37701Z supports programmable slew-rate control feature when both increasing and decreasing the output voltage, which is known as Dynamically Voltage Scaling (DVS). The ramp slew-rate can be set to 1V/mS, 2.5V/mS, 5V/mS or 10V/mS through bit1 and bit0 of control register. Moreover, the operation mode during DVS region can be adjusted through control register RPWM bit. When Ramp-PWM function is activated, the device will change to Forced PWM mode operation during DVS region and change back to auto PFM mode after reaching target output(see the Ramp-PWM Function for more information).

Ramp-PWM Function

If Ramp-PWM function is enabled, the device operates in FPWM(Forced Pulse Width Modulation) when it ramps from one output voltage to another during dynamic voltage scaling. This function is useful when you want to make sure that dynamic voltage scaling ramps the output voltage up and down in a controlled way. If the device operates in Auto PFM Mode and Ramp-PWM is enabled, the device allows both sink and source current from device output. If the device operates in Auto PFM Mode and Ramp-PWM is disabled, the devices cannot always control the ramp from a higher output voltage to a lower output voltage, because in Auto PFM or PWM Mode the devices cannot sink current from device output.

To enable Ramp-PWM function, set the RPWM bit (Address 0X01[2]) to '1'.

To disable Ramp-PWM function, clear the RPWM bit (Address 0X01[2]) to '0'.

Enable

The AWP37701Z provides an EN pin, as an external chip enable control, to enable or disable the device. If VEN is held below a logic-low threshold voltage, switching is inhibited even if the VIN voltage is above VIN undervoltage lockout threshold. During shutdown mode, the converter is disabled and all registers will reset to default value. During shutdown mode, the supply current can be reduced to I_{SD} . A discharge event happens when the EN pin pulls from high to low (see the Discharge for more information). The EN pin is compatible with 1.8V and 3.3V voltage levels.

Discharge

The discharge event happens when the EN pin pulls from high to low. A typical 200Ω resistor is connected from output to ground when discharge starts. The discharge event lasts for 20m seconds.

Over-Current Protection

The device features a cycle-by-cycle current limit circuit, which limits the peak inductor current in the event of an overload. The exact value of the maximum inductor current during an overload is determined by the inductor current rising rate (e.g., operating conditions) and the converter's response delay time (e.g., switching mode).

Input UVLO Protection

In addition to the EN pin, the AWP37701Z also provides enable control through the VIN pin. If VEN rises above V_{ENH} first, switching will still be inhibited until the VIN voltage rises above V_{UVLO} . It is to ensure that the internal regulator is ready so that operation with not-fully-enhanced internal MOSFET switches can be prevented. After the device is powered up, if the VIN voltage goes below the UVLO falling threshold voltage ($V_{UVLO} - \Delta V_{UVLO}$), this switching will be inhibited; if VIN voltage rises above the UVLO rising threshold (V_{UVLO}), the device will resume switching. During shutdown mode, the converter is disabled and all registers will reset to default value.

Over-Temperature Protection

When the junction temperature exceeds the OTP threshold value, the IC will shut down the switching operation. Once the junction temperature cools down and is lower than the OTP lower threshold, the converter will automatically resume switching. When the device detects an over-temperature condition, it asserts the TSD_STATUS bit (Address 0X02[4]) and the TSD_FLAG bit (Address 0X02[1]) to '1'. The TSD_STATUS bit (Address 0X02[4]) is automatically set to '0' when the junction temperature of the device is below 130°C. Similarly, the device clears the TSD_FLAG bit (Address 0X02[1]) to '0' upon reading the Status register.

Over-Voltage Protection

The AWP37701Z integrates an Output Ove-Voltage Protection (OVP) circuit to protect the system and downstream components from potential damage caused by excessive voltage levels. When the output voltage exceeds 5.7V, the OVP circuit is triggered, disabling the power switch to prevent damage. Once the output voltage drops below 5.4V, the OVP condition is cleared, and the power switch resumes normal operation. This ensures reliable protection of the system during overvoltage events.

Under-Voltage Protection

The AWP37701Z provides Hiccup Mode for Under-Voltage Protection (UVP). When the VOUT voltage drops below 85% of Target VOUT, the UVP function will be triggered to shut down switching operation. During this condition, the Power Good (PG) bit (Address 0X02[0]) is set to '0'. If the UVP condition remains for a period, the AWP37701Z will retry to build up output voltage automatically. When the UVP condition is removed, the converter will soft-start to target voltage and resume normal operation.

Interface

The AWP37701Z supports the I²C-Bus specification up to 1 MHz and operates in slave mode, supporting 7-bit addressing.

Register contents remain intact as long as VIN supply voltage remains above VIN undervoltage lockout threshold and EN voltage above logic-high threshold voltage. To make sure that the I²C transfer function is

correctly reset, it is recommended that the I²C master initiates a STOP condition on the I²C bus after the initial power up of SDA and SCL pullup voltages.

I²C INTERFACE

The AWP37701Z supports the I²C protocol. The maximum frequency supported by the I²C is 1MHz. The pull-up resistor for the SDA and SCL can be selected from 1kΩ to 10kΩ. Usually, 4.7kΩ is recommended for 400 kHz I²C, 1kΩ is recommended for 1MHz I²C. The voltage from 1.2V to 3.3V is allowed for the I²C interface. Additionally, the I²C device supports continuous reading and writing operations.

DEVICE ADDRESS

The I²C device address is 7-bit (A7~A1), when the VSEL pin is connected to logic low, the device address should be configured to 76h (1110110b); The VSEL pin is connected to logic high, the device address should be configured to 75h (1110101b). Followed by the R/W bit A0 (Read=1/Write=0). Set A0 to "0" for writing and "1" for reading.

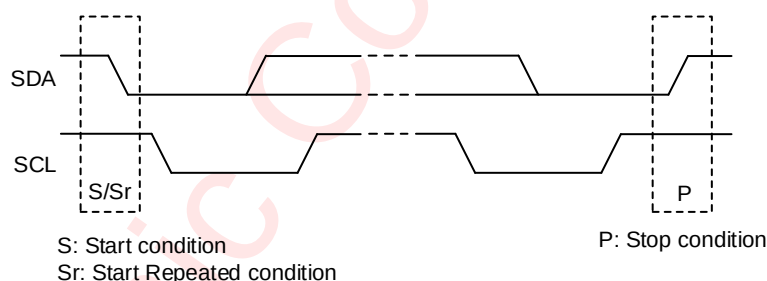
For example, The VSEL pin is connected to logic low.

A7	A6	A5	A4	A3	A2	A1	A0
1	1	1	0	1	1	0	R/W

I²C START/STOP

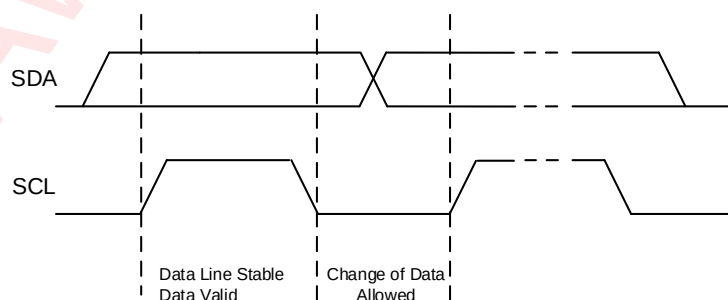
I²C Start: SDA changes from high level to low level when SCL is high level.

I²C Stop: SDA changes from low level to high level when SCL is high level.



DATA VALIDATION

When SCL is high level, SDA level must be constant. SDA can be changed only when SCL is low level.

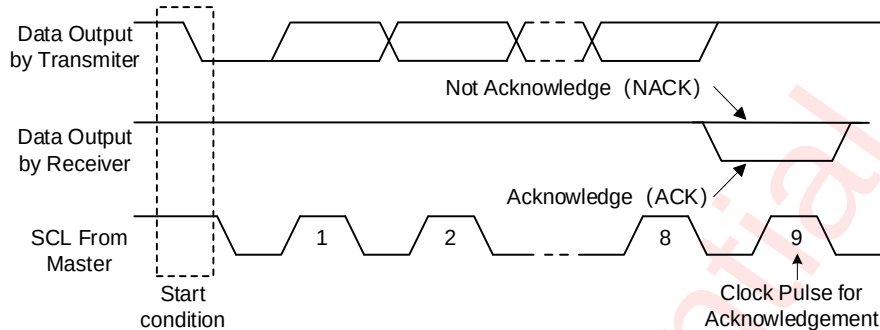


ACK (ACKNOWLEDGEMENT)

ACK means the successful transition of I²C bus data. After master sends an 8-bit data, SDA must be released;

SDA is pulled to GND by slave device when slave acknowledges.

When master reads, slave device sends 8-bit data, releases the SDA and waits for ACK from master. If ACK is sent and I²C STOP is not sent by master, slave device sends the next data. If the master sends NACK, slave device stops to send data and waits for I²C stop.



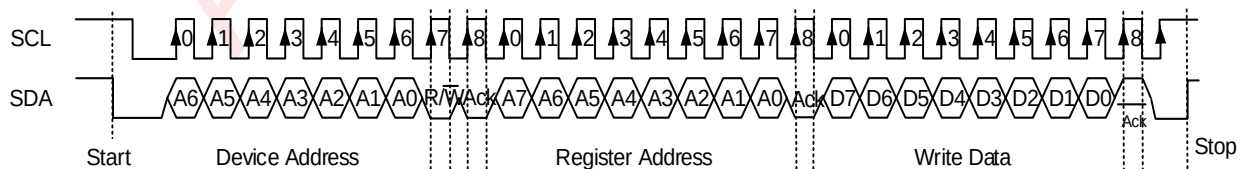
WRITE CYCLE

One data bit is transferred during each clock pulse. Data is sampled during the high state of the serial clock (SCL). Consequently, throughout the clock's high period, the data should remain stable. Any changes on the SDA line during the high state of the SCL and in the middle of a transaction, aborts the current transaction. New data should be sent during the low SCL state. This protocol allows a single data line to transfer both command/control information and data using the synchronous serial clock.

Each data transaction is composed of a start condition, a number of byte transfers (set by the software) and a stop condition to terminate the transaction. Every byte written to the SDA bus must be 8 bits long and is transferred with the most significant bit first. After each byte, an Acknowledge signal must follow.

In a write process, the following steps should be followed:

- Master device generates Start condition. The "Start" signal is generated by lowering the SDA signal while the SCL signal is high.
- Master device sends slave address (7-bit) and the data direction bit R/W = 0).
- Slave device sends acknowledge signal if the slave address is correct.
- Master sends control register address (8-bit).
- Slave sends acknowledge signal.
- Master sends data byte to be written to the addressed register.
- Slave sends acknowledge signal.
- If master will send further data bytes the control register address will be incremented by one after acknowledge signal (repeat step f and g).
- Master generates Stop condition to indicate write cycle end.

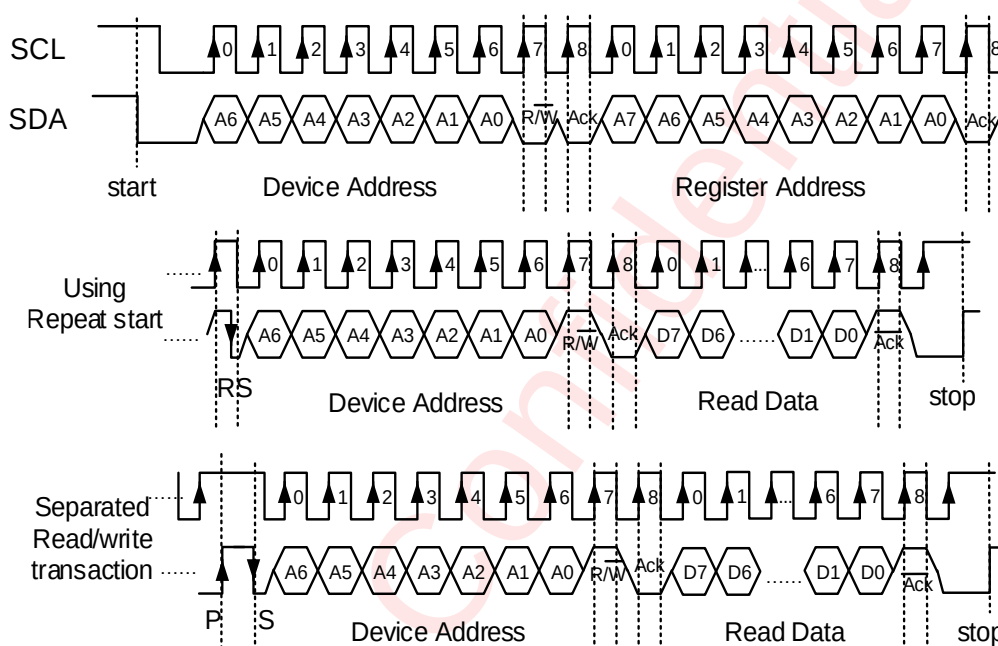


READ CYCLE

In a read cycle, the following steps should be followed:

- Master device generates Start condition.
- Master device sends slave address (7-bit) and the data direction bit (R/W = 0).

- c) Slave device sends acknowledge signal if the slave address is correct.
- d) Master sends control register address (8-bit).
- e) Slave sends acknowledge signal.
- f) Master generates Stop condition followed with Start condition or Repeat Start condition.
- g) Master device sends slave address (7-bit) and the data direction bit (R/W = 1).
- h) Slave device sends acknowledge signal if the slave address is correct.
- i) Slave sends data byte from addressed register.
- j) If the master device sends acknowledge signal, the slave device will increase the control register address by one, then send the next data from the new addressed register.
- k) If the master device generates Stop condition, the read cycle ends.



Register Description

Addr	Name	Type	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	Init
0x01	CONTROL	R/W	RESERVED		ENABLE	RESERVED	FPWM	RPWM	SLEW		0x20
0x02	STATUS	R	RESERVED	SCP_STATUS	OCP_STATUS	TSD_STATUS	SCP_FLAG	OCP_FLAG	TSD_FLAG	PG	0x00
0x03	DEVICE_IDENTITY	R	DEVICE				Manufacture ID				0x40
0x04	VOUT1	R/W	RESERVED	VOUT1							0x33
0x05	VOUT2	R/W	RESERVED	VOUT2							0x3B

Register Detailed Description

0x01: CONTROL

Bit	Symbol	Type	Description	Default
7:6	Reserved	R	Reserved	00
5	ENABLE	R/W	Control Operation of the Converter 0 = Converter operation disabled 1 = Converter operation enabled (default)	1
4	Reserved	R/W	Reserved	0
3	FPWM	R/W	Set Forced PWM Operation 0 = Disable (default) 1 = Enable	0
2	RPWM	R/W	Set Ramp PWM Operation 0 = Disable (default) 1 = Enable	0
1:0	SLEW	R/W	Set the Slew Rate. Set the slew rate of the output voltage change to a new value. 00 = 1V/ms (default) 01 = 2.5V/ms 10 = 5V/ms 11 = 10V/ms	00

0x02: STATUS

Bit	Symbol	Type	Description	Default
7	Reserved	R	Reserved	0
6	SCP_STATUS	R	The status of the short circuit Function 0 = Normal operation (default) 1 = An short circuit event is occurring	0
5	OCP_STATUS	R	The status of the Over-Current Function 0 = Normal operation (default) 1 = An over-current event is occurring	0
4	TSD_STATUS	R	The status of the Over-Temperature Function 0 = Normal operation (default) 1 = An over-temperature event is occurring	0
3	SCP_FLAG	R/C	The status of the short circuit Function It is cleared only if this register is read	0

			0 = Normal operation (default) 1 = An short circuit event was detected	
2	OCP_FLAG	R/C	The flag of the Over-Current Function It is cleared only if this register is read 0 = Normal operation (default) 1 = An over-current event was detected	0
1	TSD_FLAG	R/C	The flag of the Over-Temperature Function It is cleared only if this register is read 0 = Normal operation (default) 1 = An over-temperature event was detected	0
0	PG	R	Output Power-Good if the power-not-good condition no longer exists. 0 = A power-not-good event was detected. The output voltage is below the PG voltage threshold with a 1ms deglitch time (default) 1 = Power-good. The output voltage exceeds the PG voltage threshold with a 1ms deglitch time	0

0x03: DEVICE_IDENTITY

Bit	Symbol	Type	Description	Default
7:4	Device	R	Device ID	0100
3:0	Manufacture ID	R	Device VERSION	0000

0x04: VOUT1

Bit	Symbol	Type	Description	Default
7	Reserved	R	Reserved	0
6:0	VOUT1	R/W	VOUT1 register determines the device output voltage if the VSEL pin is logic low. $VOUT = 2.025 + (VOUT1[6:0] \times 0.025)V$ (default 3.3V)	0110011

0x05: VOUT2

Bit	Symbol	Type	Description	Default
7	Reserved	R	Reserved	0
6:0	VOUT2	R/W	VOUT2 register determines the device output voltage if the VSEL pin is logic high. $VOUT = 2.025 + (VOUT1[6:0] \times 0.025)V$ (default 3.5V)	0111011

Application Information

Typical Application

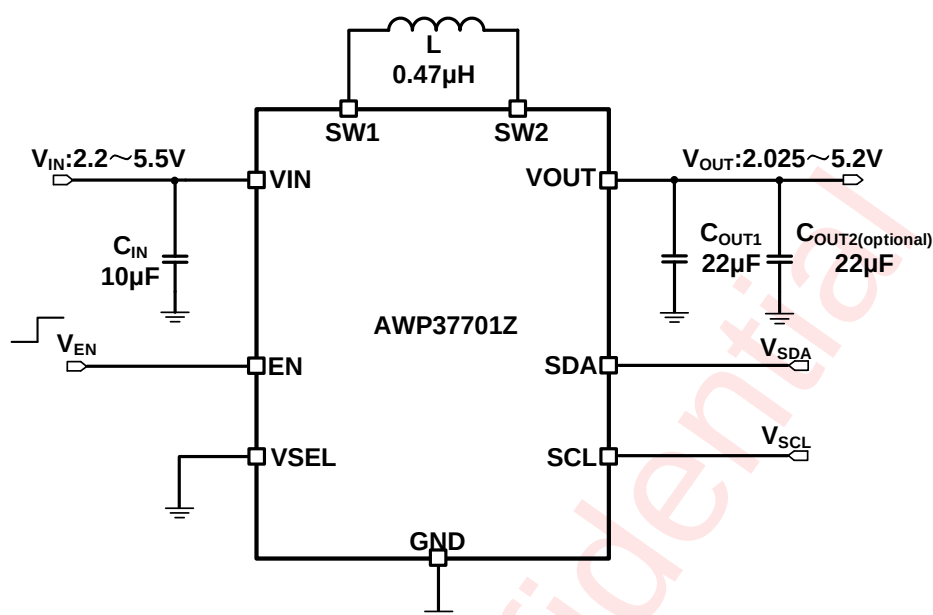


Figure 4 AWP37701Z Typical Application Circuit

NOTE: When the output current is $\leq 1A$, a $22\mu F$ COUT is recommended. When the output current is $>1A$, $2 \times 22\mu F$ COUT capacitors are recommended. The recommended maximum output current should not exceed 1.5A.

Inductor Selection

The AWP37701Z recommended nominal inductance value is $0.47\mu H$ to achieve advantage performance. For high efficiencies, use an inductor with a low DC resistance (DCR) and low core losses.

The saturation current of the inductor must be greater than the maximum inductor current in your application. To include sufficient margin for worst-case and transient operating conditions, The AWP37701Z recommends you use an inductor with saturation current that is at least 20% higher than the maximum inductor current in your application. The maximum current in the inductor occurs when the device operates in boost mode and the following is true:

The input voltage is at its minimum value.

The output voltage is at its maximum value.

The output current is at its maximum value.

To calculate the maximum inductor current, first calculate the maximum duty cycle during boost operation (which is when the maximum inductor current occurs).

$$Duty_{Max} = \frac{(V_{OUTMax} - V_{INMin})}{V_{OUTMax}}$$

Next, calculate the maximum inductor current.

$$I_{LMAX} = \frac{I_{OUTMax}}{\eta \cdot (1 - Duty_{Max})} \cdot \frac{Duty_{Max} \cdot V_{INMin}}{2 \cdot f_{SW} \cdot L}$$

To include enough margin for transient conditions, The AWP37701Z recommends you use an inductor with a saturation current rating at least 20% higher than the calculated maximum current.

Input capacitor Selection

Steady state and transient response operation performance also depend on input voltage stability or not. The AWP37701Z at least a 10μF input capacitor is recommended to prevent input voltage instability with application operation. And that suggest placed as close as possible to the VIN and GND pins of the IC is recommended. If the input supply is more than a few centimeters from the device, we recommend you add some bulk capacitance to the ceramic bypass capacitors. A 100μF electrolytic capacitor is a typical selection for the bulk capacitance.

Output capacitor Selection

The ripple voltage is an important index for choosing output capacitor. This portion consists of two parts. One is the product of ripple current with the ESR of the output capacitor, while the other part is formed by the charging and discharging process of the output capacitor. Output capacitor is selected according to output ripple which is calculated as below equation.

$$\Delta V_{OUT} = \Delta V_{ESR} + \Delta V_{OUTCAP}$$

$$\Delta V_{ESR} = I_{CRMS} \cdot R_{CESR}$$

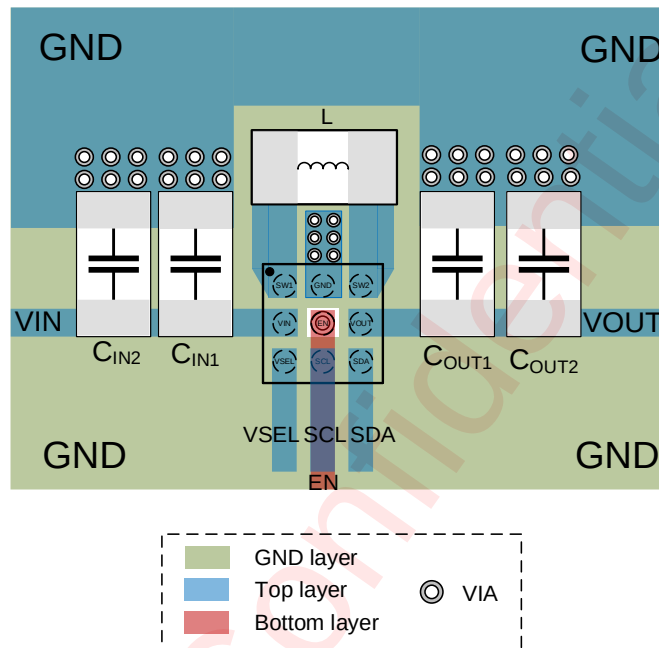
$$\Delta V_{OUTCAP} = \frac{I_{OUT} \cdot Duty}{f_{SW} \cdot C_{OUTMIN}}$$

User can use equation choose capacitor to meeting systems ripple specification. *When the IOUT is ≤1A*, at least 22μF/16V/X5R capacitors is recommended to matching application with VOUT ripple request and stability performance. *When the IOUT is >1A*, at least 2*22μF/16V/X5R capacitors are recommended to matching application with VOUT ripple request and stability performance.

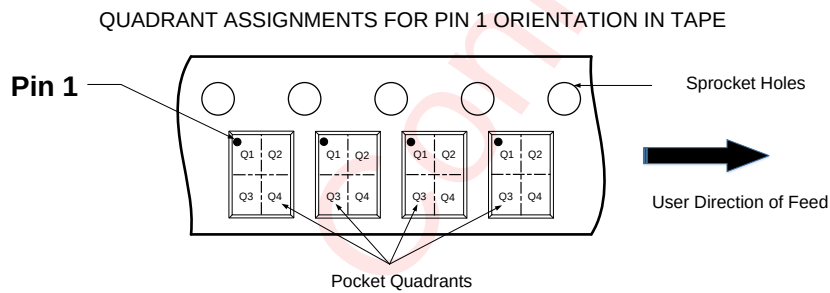
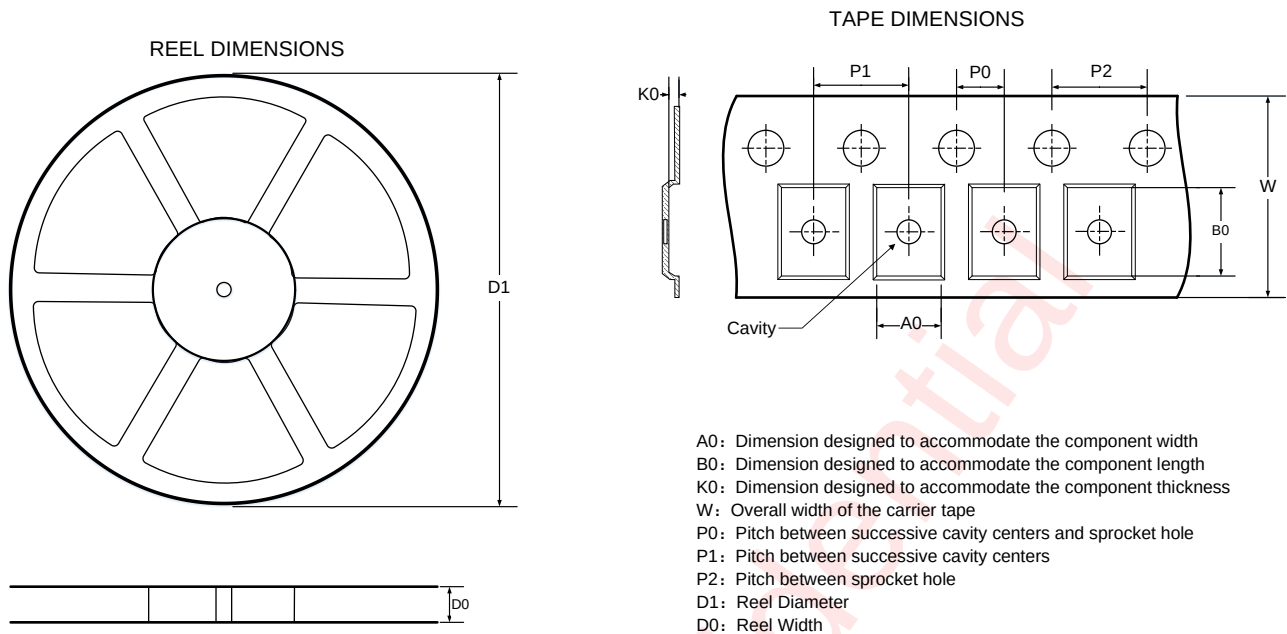
PCB Layout Consideration

The AWP37701Z is a synchronous buck-boost converter, to obtain the optimal performance, PCB layout should be considered carefully. Here are some guidelines:

1. C_{IN} , C_{OUT} should be placed as close to chip as possible.
2. Wide and short traces should be used for main current path and the power ground paths.
3. Switching node (SW1 and SW2) are with high frequency voltage swing and should be kept at small area. Keep analog components away from the SW1 and SW2 node to prevent stray capacitive noise pickup.



Tape And Reel Information



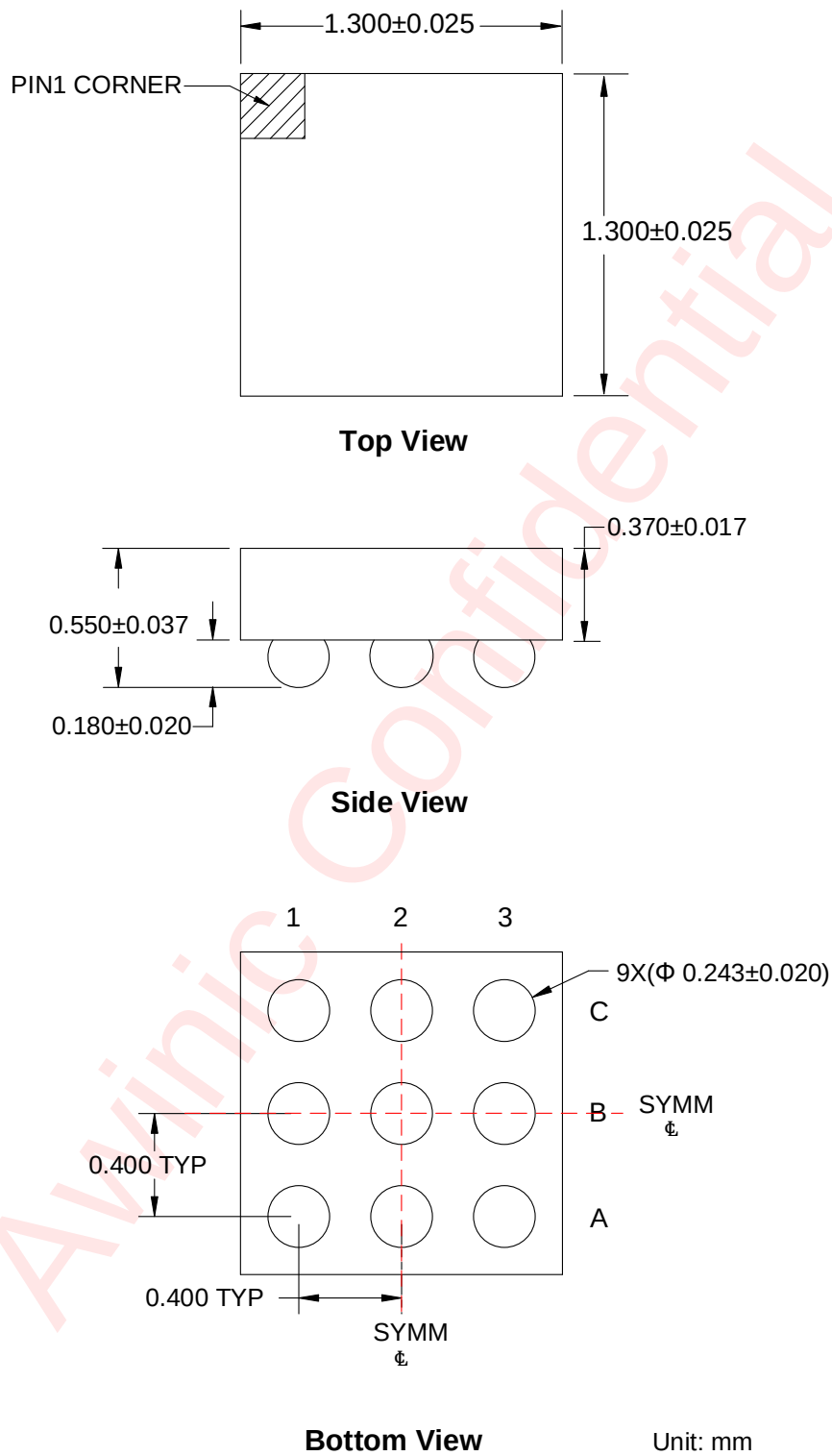
Note: The above picture is for reference only. Please refer to the value in the table below for the actual size

DIMENSIONS AND PIN1 ORIENTATION

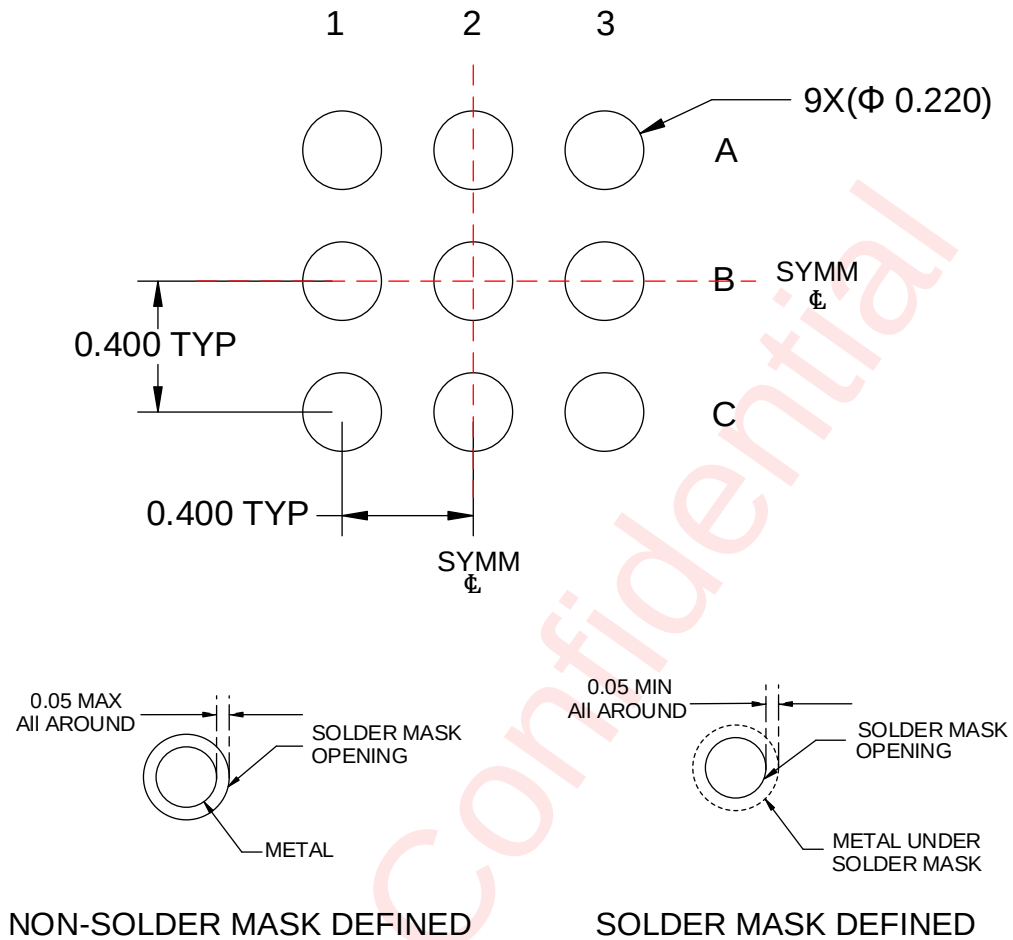
D1 (mm)	D0 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P0 (mm)	P1 (mm)	P2 (mm)	W (mm)	Pin1 Quadrant
180.00	8.60	1.50	1.46	0.68	2.00	4.00	4.00	8.00	Q1

All dimensions are nominal

Package Description



Land Pattern Data



Unit: mm

Revision History

Version	Date	Change Record
V1.0	Aug. 2025	Officially released
V1.1	Nov. 2025	Update the description of the Vout ripple operation mode in the <i>Typical Characteristics</i> section. (P10)
V1.2	Jun. 2026	1. Update Typical Application Circuit. (P2) 2. Update VSEL and EN Pin Definition. (P3) 3. Update VSEL in <i>Detailed Functional Description</i> . (P14)

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